A Thesis Report On

Impact of Capital to Risk Weighted Asset Ratio on Bank Profitability: Evidence from Bangladesh's Commercial Banks

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Letter of Transmittal

August 31, 2025

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Dear Sir,

Attached is the report as assigned, entitled Impact of Capital to Risk Weighted Asset Ratio on Bank Profitability: Evidence from Bangladesh's Commercial Banks

I am delighted to be able to prepare my paper for completing my MBA program majoring in Banking.

This report gave me the opportunity to explore the current situation of banking industry. I am thankful to you for your continuous support and patience that you have provided me throughout my thesis tenure despite your busy schedule. I am submitting my report to you for your kind assessment. I, once again, thanking you for your kind supervision, direction, communication and cooperation.

Sincerely yours,

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DECLARATION

I do hereby solemnly declare that the work presented in this thesis report has been carried out by me and has not been previously submitted to any other University/College/Organization for an academic qualification/certificate/diploma or degree.

The work I have presented does not breach any existing copyright and no portion of this report is copied from any work done earlier for a degree or otherwise. I further undertake to indemnify the Department against any loss or damage arising from breach of the foregoing obligations.

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Supervisor's Certificate

This is to certify that the Thesis Report on "Impact of Capital to Risk Weighted Asset Ratio on Bank Profitability: Evidence from Bangladesh's Commercial Banks" in the bona fide record at the report is done by Inzamam Haider Khan, ID: 1871 as a partial fulfillment of the requirement of Masters of Business Administration (MBA) degree from the Department of Banking, University of Dhaka.

The Report has been prepared under my guidance and is a record of the bona fide work carried out successfully.

…….

Md. Golam Ramij

Assistant Professor

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I consider this project a significant milestone in my career; nevertheless, despite my utmost care, limitations may have led to minor errors, for which I sincerely apologize. My appreciation also goes to everyone who willingly offered their time and expertise.

Executive Summary

This paper examines how capital adequacy shapes profitability and risk in Bangladesh's private commercial banks during a turbulent decade marked by rising NPLs, capital shortfalls, policy forbearance, and interest-rate regime shifts. Using a balanced panel of 23 PCBs over 2015–2024 (230 bank-years), we model ROA, ROE, and NPLs against CRAR, bank size (log assets), equity-to-liabilities, and macro controls (GDP growth, CPI inflation, unemployment). Estimator choice follows specification tests: random effects for ROA/ROE (Hausman fails to reject), fixed effects for NPL (xtoverid rejects RE orthogonality); all inference uses bank-clustered standard errors and diagnostics confirm serial correlation (Wooldridge) and manageable multicollinearity (mean VIF ≈ 2.6). Three results stand out. First, capital matters: higher CRAR is robustly associated with higher ROA and lower NPLs, confirming a resilience channel whereby thicker buffers sustain margins and curb credit losses; the equity-to-liabilities ratio also supports ROA, indicating value in simple (un-risk-weighted) capital alongside CRAR. ROE loads positively on CRAR but is less precise, reflecting ROE's volatility and denominator effects. Second, macro headwinds weigh on performance: unemployment and inflation compress profitability, while NPLs rise with adverse macro conditions—consistent with post-pandemic normalization and inflation shocks. Third, size is not a reliable driver once heterogeneity is controlled for, suggesting that governance and asset mix, rather than scale alone, differentiate outcomes. Trend plots corroborate the econometrics: CRAR eroded in later years as NPLs climbed, while ROA/ROE weakened sharply in 2024. Policy recommendations follow directly: (i) targeted recapitalization and credible buffers under PCA; (ii) accelerated NPL resolution (provisioning discipline, write-offs, potential asset-management vehicle); (iii) transparent leverage metrics and dividend restraint until buffers are rebuilt; (iv) risk-based pricing and better pass-through to protect margins; (v) tighter supervision and governance; (vi) bank-level stress tests linked to PCA triggers; and (vii) stronger data quality and IFRS-9/ECL practices. Limitations include secondary disclosures, possible NPL measurement inconsistencies, short time dimension (T=10), and potential endogeneity between profits and capital; future work should consider dynamic GMM, non-linear CRAR thresholds, governance and portfolio structure, market-based risk indicators, and policy event studies. Overall, the evidence points to a clear agenda: rebuild capital, resolve NPLs, and restore risk-sensitive pricing to stabilize profitability and intermediation.

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Chapter 1: Introduction

1.1 Background of the Study

The banking sector in any economy serves as the lifeblood of economic activity by facilitating the flow of credit, capital formation, and investment, and by providing a safe environment for deposits. In Bangladesh, the role of commercial banks is critical, given their direct influence on economic growth, employment, and wealth generation. However, over the past decade, Bangladesh's banking industry has encountered several challenges that have raised questions about its stability and profitability. One of the most pressing concerns is the Capital to Risk-Weighted Assets Ratio (CRAR), a key indicator that determines a bank's financial health by measuring its capital adequacy against potential risks (Bangladesh Bank, 2014).

The capital to risk-weighted assets ratio (CRAR), also referred to as the capital adequacy ratio (CAR), is a regulatory standard designed to ensure that banks can absorb losses and meet statutory capital requirements. This ratio is calculated by dividing a bank's capital by its risk-weighted assets. Under Basel III as implemented by Bangladesh Bank (BB), banks are required to hold 12.5% total capital in normal times, comprising a 10% minimum plus a 2.5% capital conservation buffer (CCB); the framework was fully phased in by 2019 (Bangladesh Bank, 2014).

with an additional capital buffer to protect against economic shocks. In Bangladesh, this regulation is enforced by the Bangladesh Bank (BB), the central bank. However, despite these regulatory frameworks, the banking sector has struggled with consistent declines in capital adequacy.

Over the last decade, the capital adequacy ratio of several banks in Bangladesh has dipped below the required threshold. The most recent figures suggest that the aggregate CRAR of the banking sector fell to 6.86% in September 2024, with 16 out of the 61 private commercial banks (PCBs) failing to meet the minimum capital adequacy requirement set by the regulatory authorities (Tonmoy Modak , 2025)This decline in CRAR coincides with an increase in non-performing loans (NPLs), which reached 20% of the total loans by the end of 2024 (Fitch Ratings, 2025). The rise in NPLs has placed a further strain on the profitability and liquidity of the banks, raising concerns about the ability of these banks to remain solvent under adverse conditions.

The relationship between capital adequacy and bank profitability is extensively documented in financial literature. A higher Capital to Risk-Weighted Assets Ratio (CRAR) is typically linked to increased profitability because it enhances a bank's resilience to economic shocks and operational risks. Conversely, excessively high CRAR may result in underutilization of capital, which can

reduce Return on Assets (ROA) or Return on Equity (ROE), both standard measures of profitability. Therefore, banks must carefully balance capital adequacy and profitability.

Bank-specific factors, including bank size and the equity-to-liabilities ratio, significantly influence financial performance. Larger banks often benefit from economies of scale, which can improve profitability and risk management. In contrast, excessive leverage or a high equity-to-liabilities ratio increases risk exposure and may negatively impact profitability.

Beyond bank-specific factors, macroeconomic variables such as GDP growth, inflation, and unemployment also have substantial effects on the banking sector. During periods of economic downturn, banks may experience lower profitability due to shrinking demand for credit, higher default rates, and rising operational costs. In contrast, economic growth often stimulates demand for loans, thereby improving profitability. Moreover, rising inflation erodes the real value of assets, and a high unemployment rate can increase the likelihood of loan defaults, both of which negatively affect profitability.

The significance of understanding these dynamics in the context of Bangladesh is heightened by the country's ongoing regulatory reforms and challenges in the banking sector. The Prompt Corrective Action (PCA) framework, adopted by Bangladesh Bank, aims to address the issue of banks failing to meet minimum CRAR requirements by enforcing corrective measures, including mergers, restructuring, and liquidation in some cases (Bangladesh Bank, 2023). Moreover, the recent liberalization of the interest rate system and the discontinuation of the SMART reference rate in 2024 have added another layer of complexity in analyzing the profitability and stability of banks (The Business Standard, 2024).

Given this background, this study aims to examine how CRAR and other related factors, such as bank size and equity/liabilities, impact profitability and risk management within the Bangladeshi banking sector. Additionally, understanding how macroeconomic variables influence these relationships during periods of stress is essential for both policymakers and banking professionals in making informed decisions (World Bank, 2025).

1.2 Problem Statement

The decline in Capital to Risk-Weighted Assets Ratio (CRAR) and the increase in Non-Performing Loans (NPLs) present substantial challenges to the profitability and stability of banks in Bangladesh. Analyzing the relationship between CRAR and profitability indicators, including Return on Assets (ROA), Return on Equity (ROE), and NPLs, is necessary to evaluate the resilience of the banking sector. Furthermore, assessing the impact of bank-specific factors such as size and equity-to-liabilities ratio, as well as macroeconomic variables including gross domestic product (GDP) growth, inflation, and unemployment, on bank profitability during periods of financial stress is critical for developing effective regulatory and managerial strategies.

1.3 Research Objectives

Broad objective

To evaluate how capital adequacy measured by the Capital to Risk-Weighted Assets Ratio (CRAR) affects bank profitability and asset quality in Bangladesh's private commercial banks (2015–2024), while accounting for bank-specific structures (size and equity-to-liabilities) and macroeconomic conditions (GDP growth, inflation, unemployment), and to translate the evidence into actionable policy guidance.

Specific objectives

1. Quantify CRAR's effect on performance and risk: Estimate the impact of CRAR on profitability (ROA, ROE) and asset quality (NPL).
2. Assess bank-level conditioning factors: Examine whether bank size (log of total assets) and the equity-to-liabilities ratio moderate the CRAR–profitability relationship (e.g., via interaction effects).
3. Identify macroeconomic influences: Evaluate how GDP growth, CPI inflation, and unemployment shape profitability and NPLs especially during stress episodes.
4. Profile bank-specific determinants: Determine the direct associations of bank size and equity-to-liabilities with ROA, ROE, and NPLs after controlling for CRAR and macro factors.

5. Provide policy recommendations: Derive evidence-based guidance for capital rebuilding, NPL resolution, and pricing/supervisory reforms (e.g., PCA implementation and rate-setting) to enhance capital adequacy and sustainable profitability.

1.4 Significance of the Study

This research is significant for several reasons:

- Policy Implications: The findings will provide valuable insights for policymakers at Bangladesh Bank to design and implement effective capital adequacy regulations and macroprudential policies.
- Bank Management: Bank managers can utilize the study's outcomes to make informed decisions regarding capital planning, risk management, and strategic planning.
- Academic Contribution: The study will contribute to the existing literature on banking profitability, particularly in the context of a developing economy like Bangladesh.
- Investor Confidence: By highlighting the factors influencing bank profitability, the research can help restore investor confidence in the banking sector.

1.5 Limitations of the Study

While this study aims to provide comprehensive insights, it is subject to certain limitations:

- Data Constraints: The analysis relies on secondary data from annual reports and regulatory filings, which may have limitations in terms of accuracy and completeness.
- Scope of Variables: The study focuses on a selected set of bank-specific and macroeconomic variables, and other factors influencing profitability may not be captured.
- Temporal Constraints: The study covers data from 2015 to 2024, and the findings may not fully account for future developments or structural changes in the banking sector.

Chapter 2: Literature Review

The Capital Adequacy Ratio (CAR), specifically the CRAR, is one of the most important indicators of a bank's financial health and resilience. A higher CRAR implies that a bank is more capable of absorbing shocks, reducing the likelihood of insolvency in the face of economic turbulence. The Basel III framework, introduced after the 2008 financial crisis, recommends that banks maintain a minimum CRAR of 10%, with an additional capital conservation buffer. Numerous studies have examined the relationship between CRAR and bank profitability. According to Akhtar et al. (2019), CRAR has a positive impact on the profitability of banks, as adequate capital enables banks to maintain stability and avoid the costs associated with financial distress. Similarly, Rahman et al. (2017) found that higher CRAR improves a bank's capacity to absorb risks, thereby positively influencing its profitability, especially during economic downturns. However, Zhao et al. (2020) argue that excessively high CRAR may reduce profitability by increasing the cost of capital, as more capital must be held, which could lead to underutilization of assets. Furthermore, Sattar et al. (2015) highlight that while CRAR ensures financial stability, it might not directly enhance profitability in the short term. This suggests that banks with higher capital ratios may be less inclined to take on risky but potentially profitable investments, thereby lowering their returns. Thus, the CRAR-profitability relationship could be non-linear, where both under-capitalization and over-capitalization could harm profitability (Molyneux & Thornton, 1992). In the context of Bangladesh, a study by Jalil et al. (2018) explored the impact of CRAR on the Return on Assets (ROA) and Return on Equity (ROE) of Bangladeshi commercial banks. The study revealed a positive relationship between CRAR and profitability, but it also noted the high level of non-performing loans (NPLs) in the sector, which undermines the effectiveness of capital adequacy in driving profitability. This suggests that in emerging markets like Bangladesh, other factors may overshadow the influence of CRAR on profitability. In addition to CRAR, several bank-specific variables influence profitability. Bank size, measured by total assets, is widely considered a determinant of bank performance. Larger banks tend to benefit from economies of scale, cost efficiencies, and better risk management, leading to higher profitability. Studies such as Goddard et al. (2011) show that large banks often outperform smaller banks due to their ability to diversify risks and manage costs more effectively. However, Berger (1995) found that the relationship between bank size and profitability is not always straightforward, with larger banks sometimes facing diminishing returns due to increased operational complexity and regulatory scrutiny. The equity-to-liabilities ratio is another important

factor influencing profitability. Kohler (2012) argues that banks with higher equity ratios tend to have lower costs of capital and are less vulnerable to financial crises, which can enhance profitability. Conversely, high leverage (lower equity-to-liabilities ratio) can lead to higher returns but increases the risk of financial instability, inflation, and unemployment, play a crucial role in shaping the performance of banks. GDP growth is generally seen as a positive indicator for banking profitability, as it increases demand for credit and reduces the likelihood of loan defaults. Studies such as Bikker & Hu (2002) have shown that during periods of economic growth, banks benefit from higher lending volumes and improved asset quality, leading to higher profitability.

Inflation, however, presents a more complex challenge for banks. Liaqat et al. (2014) argue that high inflation can erode the real value of a bank's assets and reduce the profitability of banks by increasing costs. In contrast, Borio (2006) notes that moderate inflation can be beneficial for banks if it leads to higher nominal interest rates, which increase lending margins. However, high inflation can also lead to higher operational costs, and in extreme cases, it can reduce the purchasing power of consumers, leading to higher loan defaults. The unemployment rate is another macroeconomic variable that impacts banking performance. Roodman (2009) posits that an increase in unemployment negatively affects bank profitability, as it typically leads to higher loan defaults. The literature suggests that banks in high-unemployment environments face increased credit risk, which erodes profitability. In the context of Bangladesh, Hossain & Ahmed (2016) argue that macroeconomic volatility, particularly during periods of political instability, significantly affects the profitability of banks. They found that macroeconomic instability led to a reduction in loan disbursements and an increase in NPLs, both of which negatively impacted profitability. While the relationship between CRAR, profitability, and macroeconomic factors has been extensively studied globally, there is a clear gap in the literature regarding the Bangladeshi banking sector. Much of the existing research on CRAR and profitability focuses on developed economies, and little attention has been paid to how CRAR affects Bangladeshi banks amidst unique challenges such as high inflation, political instability, and a rapidly growing economy. Moreover, most studies have focused on linear relationships between CRAR and profitability, overlooking potential non-linear effects. The role of bank-specific factors such as bank size and equity-to-liabilities ratios, and their interaction with macroeconomic variables like GDP growth, inflation, and unemployment, is an area that requires further exploration in emerging market contexts like Bangladesh.

Chapter 3: Methodology

3.1 Research Design

This research employs a quantitative design and analyzes secondary data from 23 Private Commercial Banks (PCBs) in Bangladesh collected over a 10-year period (2015 to 2024). A panel data regression model is used to examine the relationships between the dependent and independent variables. Panel data regression is selected because it incorporates both time-series and cross-sectional dimensions, which enhances the robustness of the findings relative to analyses based solely on cross-sectional or time-series data.

The analysis focuses on understanding the effect of bank-specific variables and macroeconomic factors on the profitability of banks in Bangladesh. Specifically, the Capital to Risk-Weighted Assets Ratio (CRAR) is the key independent variable, with profitability measures like Return on Assets (ROA) and Return on Equity (ROE) serving as dependent variables.

3.2 Data Collection Methods

The data for this study is primarily secondary data, collected from various reliable sources:

- Annual Reports of the banks for the years 2015-2024.
- Bangladesh Bank's official publications and financial stability reports.
- World Bank Database (WDI) for macroeconomic data such as GDP growth, Inflation (CPI), and Unemployment Rate.

The banks chosen for the study include 23 Private Commercial Banks (PCBs). These banks were selected based on their data availability, as only banks with at least 10 years of operational history were included. Banks that began operations after 2015 were excluded, ensuring that the sample is consistent over the 10-year period.

The total sample size is therefore 23 banks × 10 years = 230 observations. This sample size provides a robust dataset for statistical analysis and allows for the examination of long-term trends in banking profitability and capital adequacy.

3.3 Sample Techniques

The study employs a **purposive sampling** technique, where banks were selected based on specific criteria:

1. The bank must have been operational for at least 10 years (i.e., started before 2015).
2. Data for key financial variables (CRAR, ROA, ROE, NPLs, and others) must be publicly available for the entire period from 2015 to 2024.

The justification for the sample size of 23 PCBs is based on the fact that Bangladesh has a total of 47 private commercial banks, but only a subset of these banks has a consistent historical record of relevant financial data spanning 10 years. The selection of 23 banks ensures that the data is representative of the broader banking sector, while excluding banks with limited data availability. This methodology provides a sample that is both manageable and robust, reflecting the overall state of the private banking sector in Bangladesh.

3.4 Variables and Hypothesis Statements

This study uses several variables that are classified as dependent, independent, and control variables.

3.4.1 Dependent Variables

1. Return on Assets (ROA): ROA is the primary measure of profitability in this study. It is calculated as:

   $$ROA = \text{Net Income Total}/\text{Assets} \times 100$$

   This metric measures a bank's efficiency in generating profits from its assets.

2. Return on Equity (ROE): ROE is another profitability measure that reflects how well the bank uses shareholders' equity to generate profits:

   ROE=Net Income/Shareholder's Equity×100

   Both ROA and ROE will help capture different dimensions of profitability.

3. Non-Performing Loans (NPL): While not directly used as a dependent variable in the primary models, NPL will be explored as an alternative outcome, as it provides insights into the asset quality of banks and their profitability. It also indicates the riskiness of the bank's position compared to CRAR.

   NPL=Total Classified Loans/Total Loans and Advances ×100

3.4.2 Independent Variables

1. The Capital to Risk-Weighted Assets Ratio (CRAR) serves as the primary independent variable in this analysis. CRAR is defined as the ratio of a bank's capital to its risk-weighted assets.

   CRAR=Bank's Capital/Risk-Weighted Assets (RWA)

   This ratio reflects a bank's ability to absorb losses and continue operations under financial stress.

1. Total Asset (Bank Size): Bank size, measured as the natural logarithm of total assets, is used to assess the economies of scale that large banks may enjoy, potentially leading to higher profitability.
2. Equity to Total Liabilities Ratio: This ratio captures the leverage of a bank. A higher ratio indicates lower risk and may contribute to higher profitability, as more capital buffers protect the bank from financial instability.

3.4.3 Macroeconomic Control Variables

1. GDP Growth: GDP growth is expected to have a positive impact on bank profitability, as economic expansion typically drives higher credit demand and loan repayment capacity.
2. Inflation (CPI): Inflation can affect the real value of assets and liabilities, potentially eroding profitability. This relationship may be negative, particularly when inflation is high and unanticipated.
3. Unemployment Rate: A higher unemployment rate can negatively affect bank profitability by increasing loan defaults, as fewer people can meet their financial obligations.

Table 1:Variable Details and References

Variables	Acronym	Nature of Variables	Formula	References
Return on Assets	ROA	Dependent	$ROA = \frac{Net\ Income}{Average\ Total\ Asset}$	(Business Perspectives; Robin et al., 2018), (Lee & Hsieh, 2013), (Akther, 2023)
Return on Equities	ROE	Dependent	$ROE = \frac{Net\ Income}{Shareholders' Equity}$	(Business Perspectives; Robin et al., 2018), (Lee & Hsieh, 2013)
Non-Performing Loan Ratio	NPL	Dependent	$NPL = \frac{Non\ Performing\ Loans}{Total\ Loans}$	(Akhter, 2023; Anjum, 2024)
Capital to Risk Weighted Asset Ratio	CRAR	Independent (Primary treatment)	$CRAR = \frac{Total\ Capital(Tier\ 1 + Tier\ 2)}{Risk - Weighted\ Assets}$	(Business Perspectives; Robin et al., 2018), (Akther, 2023)

Bank Size	-	Independent (Bank-specific controls)	$Bank\ Size = \ln(Total\ Assets)$	(Lee & Hsieh, 2013), (Lee & Hsieh, 2013)
Equities To Total Liabilities	-	Independent (Bank-specific controls)	$Equities\ to\ total\ Liabilities = \frac{Total\ Equities}{Total\ Liabilities}$	(Lee & Hsieh, 2013)
Inflation	-	Independent (Macroeconomic controls)	World Bank Database (WDI) for macroeconomic data	(Akhter, 2023; Anjum, 2024)
Unemployment Rate	-	Independent (Macroeconomic controls)	World Bank Database (WDI) for macroeconomic data	(Akhter, 2023; Anjum, 2024)
Gross Domestic Product Growth	GDP Growth	Independent	World Bank Database (WDI) for macroeconomic data	(Akhter, 2023; Anjum, 2024)

3.5 Hypothesis Statements

Based on the variables outlined above, the following hypotheses are formulated:

H1: There is a positive relationship between CRAR and ROA/ROE.

Justification: Higher CRAR indicates better capital adequacy, which can lead to improved bank profitability.

H2: Higher CRAR is associated with lower NPLs.

Justification: A higher CRAR provides banks with a buffer to absorb losses, reducing the likelihood of non-performing loans.

H3: Bank size (Total Assets) has a positive relationship with ROA/ROE.

Justification: Larger banks achieve higher profitability due to economies of scale, enhanced risk management practices, and diversified portfolios, which collectively improve operational efficiency and financial performance.

H4: Equity to Liabilities Ratio is positively associated with ROA/ROE.

Justification: Banks with higher equity relative to liabilities are less risky and have better profitability potential.

H5: GDP Growth positively influences ROA/ROE.

Justification: Higher economic growth generally increases demand for credit and reduces loan defaults, enhancing profitability.

H6: Inflation (CPI) negatively affects ROA/ROE.

Justification: High inflation erodes the real value of assets and may increase operational costs, reducing profitability.

H7: Unemployment Rate negatively influences ROA/ROE.

Justification: A higher unemployment rate increases the risk of loan defaults, thus lowering profitability.

3.6 Analytical Techniques

The primary analytical technique used in this study is panel data regression. Panel data allows for the inclusion of both cross-sectional (banks) and time-series (years) data, providing more comprehensive insights into the variables' relationships. The study will employ Fixed Effects (FE) and Random Effects (RE) models to capture unobserved heterogeneity across banks. Additionally, robust standard errors will be used to account for potential heteroskedasticity.

3.7 Economic Model

The relationship between the Capital to Risk-Weighted Assets Ratio (CRAR) and bank profitability, measured by Return on Assets (ROA) and Return on Equity (ROE), will be examined using an econometric model. This model will test the hypotheses outlined in Section 3.5 by modeling profitability as a function of bank-specific factors and macroeconomic conditions.

The basic panel data regression model for this study is as follows:

3.7.1 Basic Model Specification

$\text{Profitability}_{it}=\alpha_i+\beta_1\text{CRAR}_{it}+\beta_2\text{Total Assets}_{it}+\beta_3\text{Equity-to-Liabilities}_{it}+\gamma_1\text{GDP Growtht}+\gamma_2\text{Inflation}_t+\gamma_3\text{Unemployment}_t+\epsilon_{it}$

Where:

- Profitability (Dependent Variables):
  - ROA (Return on Assets)
  - ROE (Return on Equity)

These are used interchangeably as the dependent variables in the model to analyze the profitability of banks.

- CRAR (Independent Variable):
  - The key independent variable, Capital to Risk-Weighted Assets Ratio (CRAR), reflects a bank's capital adequacy.
  - β_1 measures the effect of CRAR on profitability.
- Bank-Specific Control Variables:
  - Total Assets: A proxy for bank size; measured as the natural logarithm of total assets (ln (Total Assets))
  - Equity-to-Liabilities: A measure of a bank's leverage, indicating how much equity the bank holds relative to its liabilities.
  - β_2 and β_3 capture the effect of these variables on profitability.
- Macroeconomic Control Variables:
  - GDP Growth: A macroeconomic variable representing the overall economic growth, which is expected to positively affect bank profitability.
  - Inflation (CPI): The rate of inflation, expected to negatively impact bank profitability by eroding the real value of assets and increasing costs.
  - Unemployment Rate: The rate of unemployment in the economy, expected to have a negative effect on bank profitability by increasing loan defaults.

- γ_1, γ_2, and γ_3 represent the coefficients for these macroeconomic variables.

- Error Term (ϵ_{it}): This term captures any unobserved factors or random noise that might influence the profitability of banks, beyond what is explained by the model's independent variables.
- Fixed Effects (α_i): The model includes bank-specific fixed effects (α_i) to control for unobserved heterogeneity across banks. This accounts for any time-invariant characteristics that differ across banks, such as management practices, bank-specific governance, or historical reputation.

3.7.2 Extended Model with NPLs as a Dependent Variable

An alternative model includes NPLs (Non-Performing Loans) as a dependent variable to examine the impact of CRAR on asset quality:

$$NPL_{it}=\alpha_i+\beta_1 CRAR_{it}+\beta_2 \text{Total Assets}_{it}+\beta_3 \text{Equity-to Liabilities}_{it}+\gamma_1 \text{GDP Growth}_t+\gamma_2 \text{Inflation}_t+\gamma_3 \text{Unemployment}_t+\epsilon_{it}$$

Where NPL measures the percentage of loans that are overdue or in default. This model allows us to see how CRAR affects the risk profile of banks, as well as how profitability (ROA/ROE) is impacted by the quality of the bank's loan book.

3.8 Justification for the Model

The proposed model is designed to address the primary research questions:

CRAR's Impact on Profitability: The inclusion of CRAR as the primary independent variable allows us to test the hypothesis that higher capital adequacy leads to improved profitability (ROA/ROE) and lower NPLs, especially during economic stress periods.

Influence of Bank-Specific Factors The bank size (measured by total assets) and leverage ratio (equity-to-liabilities) provide insight into how individual banks' operational scale and capital structure influence profitability. Larger banks tend to be more diversified and better positioned to weather economic fluctuations.

Macroeconomic Factors: By incorporating macroeconomic controls like GDP growth, inflation, and unemployment, the model accounts for the broader economic conditions that might impact profitability across the banking sector.

Non-Performing Loans as a Risk Indicator: Including NPLs as a dependent variable in an alternative model allows for an exploration of bank risk. A higher CRAR is expected to mitigate the impact of economic shocks and improve asset quality, thus lowering NPLs.

3.9 Statistical Methods

 Panel Data Regression: The primary analysis will employ Fixed Effects (FE) regression, which controls for time-invariant unobserved heterogeneity across banks. The Random Effects (RE) model will also be considered, and the choice between FE and RE will be determined using the Hausman test. This test helps identify whether the fixed effects model or the random effects model provides a better fit for the data.

 Robust Standard Errors: To account for heteroskedasticity and autocorrelation, robust standard errors will be used in all regression models. This ensures that the results are not biased due to the presence of variance in the error terms.

 Diagnostics: Various diagnostic tests will be conducted, including tests for multicollinearity (e.g., Variance Inflation Factor - VIF), serial correlation, and the Wooldridge test will be used for serial correlation. We assessed first-order serial correlation in panel errors using the Wooldridge test. For all outcomes ROA, ROE, and NPL the null of no AR (1) was rejected (Table A.D1). Accordingly, all preferred specifications report bank-clustered (panel-robust) standard errors, and we verify robustness with GLS variants (Appendix A.D2). We assessed multicollinearity among the regressors using variance inflation factors (VIF) computed from a pooled OLS with the same RHS used in the panel models (dummies excluded). All VIFs are below the conventional thresholds (mean VIF = 2.62; max = 4.09 for Equity/Total Liabilities), indicating no severe multicollinearity. Accordingly, we retain the full regressor set and report cluster-robust standard errors.

Chapter 4: Data Analysis and Results

4.1 Overview of the Data

In this section, we present the results of the empirical analysis, based on the panel data regression models discussed in Chapter 3. The data spans from 2015 to 2024 for 23 Private Commercial Banks (PCBs) in Bangladesh, totaling 230 observations. The analysis includes both bank-specific variables and macroeconomic factors as independent variables, with Return on Assets (ROA), Return on Equity (ROE), and Non-Performing Loans (NPLs) as dependent variables.

The data was collected from annual reports, Bangladesh Bank publications, and the World Bank Database (WDI). For this chapter, the main focus is on presenting the findings from the regression models and interpreting their significance in the context of the hypotheses formulated in Chapter 3. VIF statistics (mean 2.62; all <5) suggest collinearity does not materially inflate standard errors; results are robust to alternating the two capital measures (see Appendix A.C1).

4.2 Descriptive Statistics

Before proceeding with the regression results, we present the descriptive statistics for the key variables in the model. This section includes the mean, standard deviation, minimum, and maximum values of each variable.

Table 2: Statistical Summary of all variables:

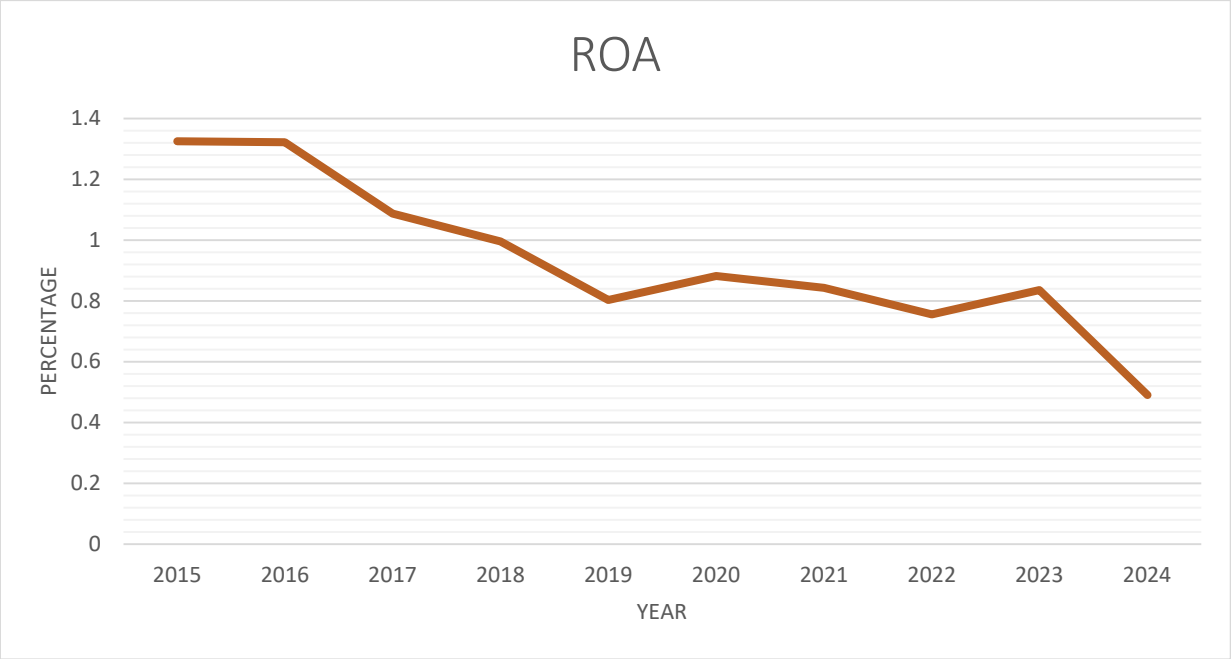
Variable	Observations	Mean (%)	Std. Dev.	Min (%)	Max (%)
ROA	230	.9335652	.6554604	-4.54	3.81
ROE	230	10.46965	10.51447	-131.14	26.1
NPL	230	5.920087	6.661806	0	67.14
CRAR	230	14.56572	3.273236	3.47	35.26
Total Asset	230	5.320123	.3805426	4.066475	5.968545
Equity To Total Liabilities	230	9.264106	3.924478	1.166203	30.29057
GDP Growth	10	6.29	1.431743	3.4	7.9
Inflation	10	6.782843	1.909494	5.513526	10.46575
Unemployment Rate	10	4.646	.4032424	4.35	5.44

Source 1: Author's Own Calculation.

Table 4.1 summarizes the distribution of key variables for 23 private commercial banks over 2015–2024 (N = 230 bank-years). Average profitability is modest: mean ROA is 0.93% with meaningful dispersion (sd 0.66pp), while ROE averages 10.47% but exhibits heavy tails (sd 10.51pp; min −131.14%), consistent with occasional large losses and thin equity bases. Asset-quality risk is elevated and skewed (mean NPL 5.92%, sd 6.66pp; max 67.14%), signaling distressed episodes in specific bank-years. Capital buffers vary widely (mean CRAR 14.57%; min 3.47%; max 35.26%), implying both regulatory shortfalls and very high capitalization in the cross-section. Bank size (log assets) spans ~1.90 log points, implying the largest bank is roughly 6.7× the smallest; the equity-to-liabilities ratio also varies substantially (mean 9.26%; range 1.17–30.29%). Macroeconomic conditions show moderate variation in GDP growth and inflation but limited movement in unemployment.

4.3 Trend Analysis

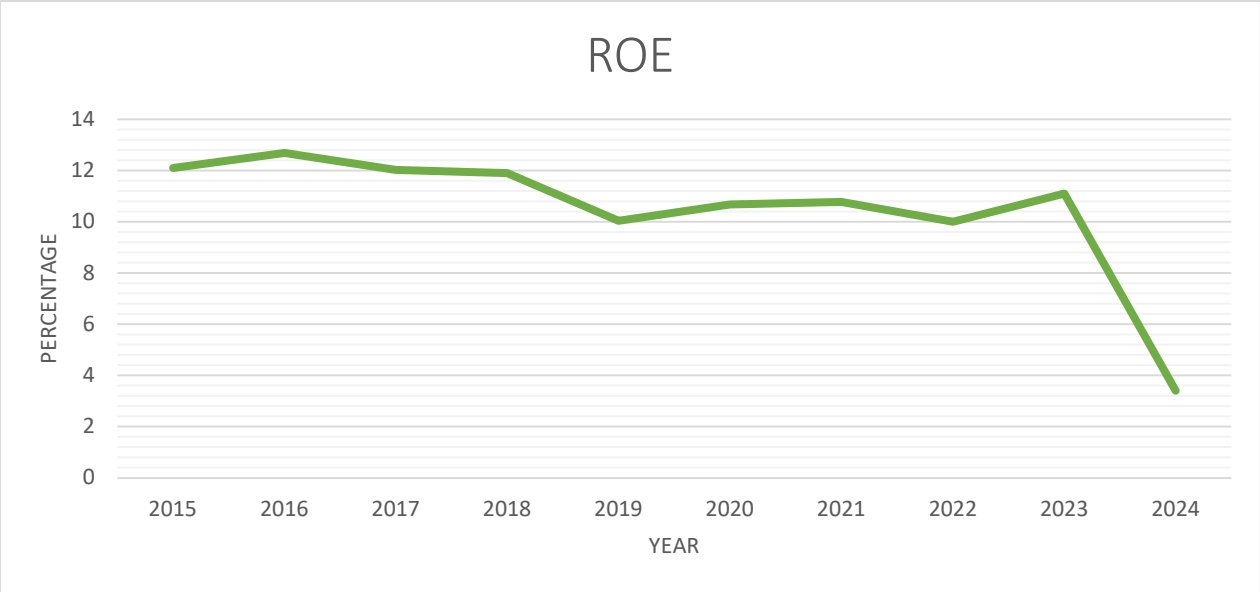
Figure 1: Ten years average of Return on Assets|



Source 2: Author's Own Calculation

ROA trends down from about 1.3% in 2015–16 to roughly 0.8% by 2019, indicating a steady squeeze on bank margins even before the pandemic. A small bounce in 2020–21 likely reflects COVID-era policy forbearance and liquidity support that masked underlying asset-quality stress and deferred provisioning. From 2022, profitability weakens again as inflation spikes, foreign-exchange pressures and import curbs raise funding and operating costs while loan demand/repayment quality softens. The modest uptick in 2023 is short-lived: pricing remains constrained coming out of years of rate caps, and rising NPLs/provisioning continue to weigh on returns. The sharp drop in 2024 is consistent with tighter liquidity and higher funding costs as rates reprice (post-cap), alongside continued asset-quality deterioration—mirroring our regression finding that macro headwinds (inflation, unemployment) and weaker capital positions compress ROA.

Figure 2:Ten years average of Return on Equity

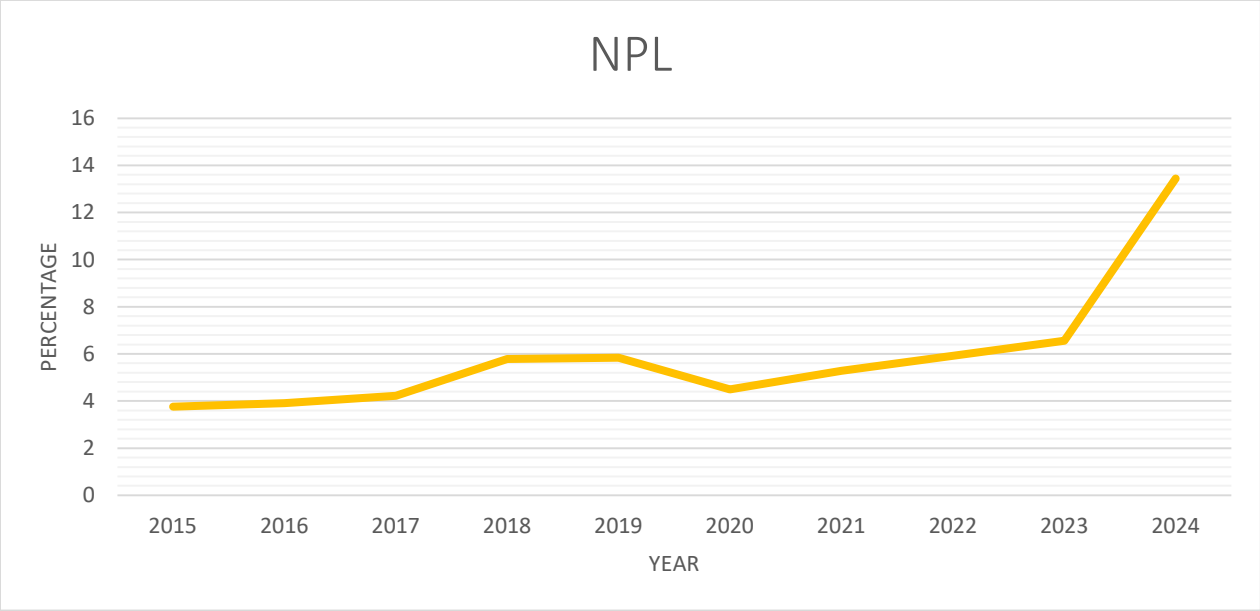


Source 3: Author's Own Calculation

ROE holds near 12% through 2015–2018, then slips to about 10% in 2019 as margin pressure and rising credit risk start to bite. A mild rebound in 2020–2021 likely reflects COVID-era forbearance and liquidity support that propped up earnings and delayed loss recognition. From 2022, high inflation, FX pressure, and import controls strain funding costs and loan performance, keeping ROE around 10% despite policy support and selective repricing. The uptick in 2023 suggests

temporary relief as activity normalizes, but provisioning needs and weaker asset quality still cap returns. The sharp fall in 2024 points to delayed loss recognition and tighter funding conditions after rate-regime changes, with heavier provisioning on rising NPLs eroding equity returns—consistent with our regressions showing macro headwinds (inflation, unemployment) significantly depress profitability and CRAR effects on ROE are imprecise.

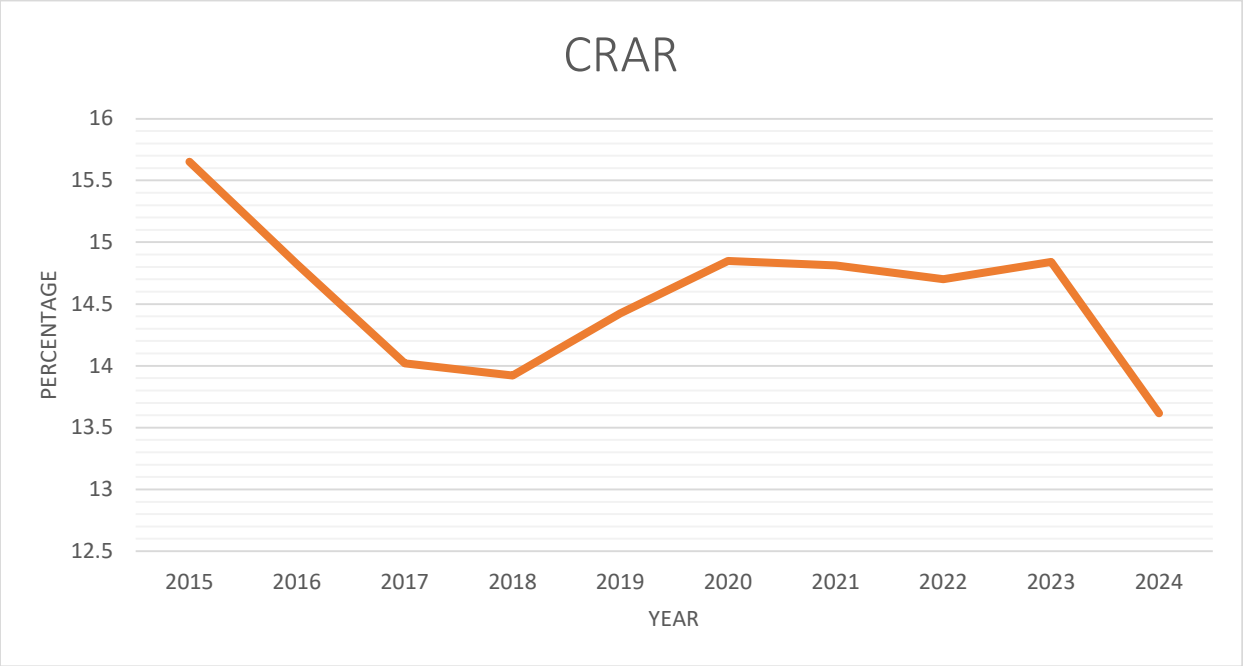
Figure 3: Ten years average of Non-Performing Loans



Source 4: Author's Own Calculation

NPLs are low and relatively stable through 2015–2017, then rise toward 6% by 2018–2019 as legacy problem loans surface and credit quality weakens. They dip in 2020—a likely effect of COVID-era forbearance, repayment moratoria, and relaxed classification rules that temporarily suppressed measured defaults. From 2021–2023, NPLs creep up again amid high inflation, FX pressure, and import restrictions that strain borrowers' cash flows and banks' recovery efforts. The sharp spike in 2024 points to delayed loss recognition as forbearance unwinds and funding costs reprice after rate-regime changes, exposing underlying asset-quality stress This trajectory matches our econometric results: lower CRAR is associated with lower NPLs, while macro headwinds (inflation, unemployment, growth shocks) raise NPLs, underscoring the need for recapitalization and faster resolution.

Figure 4: Ten years average of Capital to Risk-Weighted Asset Ratio



Source 5: Author's Own Calculation

CRAR starts high (≈15.7% in 2015) and falls to around 14% by 2017–2018, consistent with rapid balance-sheet growth and legacy NPLs absorbing capital buffers. It then rebounds in 2019–2021 toward ~14.8–14.9%, likely reflecting retained earnings, selective recapitalization, and COVID-era regulatory forbearance that eased provisioning pressure. From 2022, inflation, FX strain, and tighter liquidity raise risk-weighted assets and squeeze margins, nudging CRAR lower again. A small uptick in 2023 suggests temporary relief from partial loan repricing and balance-sheet clean-ups, but buffers remain thin the sharp drop in 2024 (~13.6%) signals renewed capital stress as higher provisioning and post-cap rate normalization erode capital—aligning with our findings that lower CRAR coincides with higher NPLs and weaker ROA, heightening system vulnerability.

4.4 Correlation Matrix

A correlation matrix is provided to examine the relationships between the independent and dependent variables. The correlation matrix helps in identifying potential multicollinearity issues and provides insights into the strength and direction of relationships between variables.

Table 3: Correlation Matrix involving all variables.

	ROA	ROE	NPL	CRAR	Total Asset	Equity To Total Liabilities	GDP Growth	Inflation	Unemployment Rate
ROA	1.000								
ROE	0.749	1.000							
NPL	-0.696	-0.723	1.000						
CRAR	0.618	0.300	-0.473	1.000					
Total Asset	-0.398	-0.070	0.259	-0.497	1.000				
Equity To Total Liabilities	0.633	0.192	-0.348	0.780	-0.699	1.000			
GDP Growth	0.142	0.121	-0.171	0.018	-0.170	0.109	1.000		
Inflation	-0.231	-0.168	0.315	-0.043	0.284	-0.209	-0.439	1.000	
Unemployment Rate	-0.132	-0.043	0.008	0.023	0.168	-0.123	-0.552	-0.129	1.000

Source 6: Author's Own Calculation.

The matrix shows patterns that are broadly consistent with banking theory. Profitability measures are tightly linked: ROA–ROE = 0.75, indicating they capture similar performance dimensions. Asset quality is inversely related to profitability: NPL correlates negatively with ROA (−0.70) and

ROE (−0.72), suggesting higher default risk coincides with lower returns. Capital strength aligns with better performance and lower risk: CRAR correlates positively with ROA (0.62) and negatively with NPL (−0.47); the equity-to-liabilities ratio shows a similar pattern (ROA = 0.63; NPL = −0.35). Bank size (log assets) is negatively related to capital ratios (CRAR = −0.50; equity/liabilities = −0.70) and to ROA (−0.40), implying larger banks in this sample tend to operate with thinner capital buffers and slightly lower asset returns. Macroeconomic conditions exhibit expected signs but modest magnitudes: GDP growth is weakly positive with profitability (ROA 0.14; ROE 0.12) and negative with NPL (−0.17), while inflation is negative for ROA (−0.23) and positive for NPL (0.32). Unemployment shows small negative links to profitability (ROA −0.13; ROE −0.04).

4.5 Regression Results

This section presents the results of the panel data regression models for both ROA, ROE and NPL as dependent variables. Each model will be presented along with the coefficients, standard errors, t-statistics, p-values, and 95% confidence intervals.

Hausman model-choice summary (FE vs RE)

Table 4: Hausman Test Output Summary for all variables.

DV	df	χ^2	p-value	Decision
ROA	6	5.06	0.537	RE
ROE	6	2.11	0.909	RE
NPL	4*	12.498*	0.014*	FE

Source 7: Author's own calculation.

*NPL decision based on the xtoverid Sargan–Hansen test after RE (df=4, χ^2=12.498, p=0.014), which rejects RE orthogonality ⇒ prefer FE

4.5.1 ROA Model Results

Model 1: Random Effects Regression for ROA

$ROA_{it}=\alpha_i+\beta_1 CRAR_{it}+\beta_2 \text{Total Assets}_{it}+\beta_3 \text{Equity-to-Liabilities}_{it}+\gamma_1 \text{GDP Growth}_t+\gamma_2 \text{Inflation}_t+\gamma_3 \text{Unemployment}_t+\epsilon_{it}$

Table 5: Random-effects GLS results for ROA (bank-clustered SEs)

Variable	Coefficient	Std. Error	z	p-value
CRAR (pp)	0.0948	0.0307	3.09	0.002***
Total assets (log)	0.1106	0.1661	0.67	0.505
Equity / Total liabilities (pp)	0.0364	0.0162	2.25	0.024**
GDP growth (pp)	−0.0593	0.0249	−2.38	0.017**
Inflation, CPI (pp)	−0.0964	0.0330	−2.92	0.003***
Unemployment rate (pp)	−0.3921	0.0995	−3.94	0.000***
Constant	1.4753	0.9285	1.59	0.112

Source 8: Author's own calculation.

Random-effects GLS with bank-clustered SEs (23 banks; 230 bank-years, 2015–2024). The model is jointly significant (Wald χ^2 (6) =218.35, $p<0.001$) and fits very well: R^2 within=0.909, R^2 between=0.928, R^2 overall=0.916. About 36% of unexplained variance is attributable to persistent bank effects (ρ=0.360). Capital matters: a 1-ppt increase in CRAR is associated with +0.095 pp higher ROA (p=0.002); a 1-ppt rise in Equity/Total Liabilities adds +0.036 pp (p=0.024). Size (log assets) is not significant. Macro headwinds reduce profitability: GDP growth −0.059 pp (p=0.017), inflation −0.096 pp (p=0.003), and unemployment −0.392 pp ($p<0.001$). Coefficients are in percentage points; ***, ** indicate significance at 1% and 5%, respectively.

4.5.2 ROE Model Results

Model 2: Random Effects Regression for ROE

$ROE_{it}=\alpha_i+\beta_1 CRAR_{it}+\beta_2 \text{Total Assets}_{it}+\beta_3 \text{Equity-to-Liabilities}_{it}+\gamma_1 \text{GDP Growth}_t+\gamma_2 \text{Inflation}_t+\gamma_3 \text{Unemployment}_t+\epsilon_{it}$

Table 6: Random-effects GLS results for ROE (bank-clustered SEs)

Variable	Coefficient	Std. Error	z	p-value
CRAR (pp)	1.3848	0.9667	1.43	0.005***
Total assets (log)	4.6511	4.0087	1.16	0.246
Equity / Total liabilities (pp)	−0.3625	0.3612	−1.00	0.316
GDP growth (pp)	−0.3655	0.2700	−1.35	0.176
Inflation, CPI (pp)	−1.5415	0.8549	−1.80	0.071*
Unemployment rate (pp)	−4.3248	1.7723	−2.44	0.015**
Constant	1.7609	22.2763	0.08	0.937

Source 9: Author's own calculation.

Random-effects GLS (bank-clustered SEs; 23 banks, 230 bank-years, 2015–2024). The model is marginally significant overall (Wald χ^2 (6) =18.69, p=0.098). Fit is high in-sample (R^2within=0.812; R^2between=0.899; R^2overall=0.851), with about 17% of residual variance between banks (ρ=0.167). CRAR is positively signed (β=1.3848, p=0.005). Size (log assets) and equity/liabilities are not significant. Macroeconomic headwinds depress ROE: unemployment is strongly negative (β=−4.3248, p=0.015), and inflation is marginally negative (β=−1.5415, p=0.071); GDP growth is negative but not significant. Coefficients are in percentage points.

4.5.3 NPL Model Results (Alternative Dependent Variable)

Model 3: Fixed Effects Regression for NPL

$$NPL_{it}=\alpha_i+\beta_1 CRAR_{it}+\beta_2 \text{Total Assets}_{it}+\beta_3 \text{Equity-to-Liabilities}_{it}+\gamma_1 \text{GDP Growth}_t+\gamma_2 \text{Inflation}_t+\gamma_3 \text{Unemployment}_t+\epsilon_{it}$$

Table 7:. Fixed-effects results for NPL, bank-clustered SEs

Variable	Coefficient	Std. Error	t	p-value
CRAR (pp)	−0.1212	0.0335	−3.62	0.002***
Total assets (log)	0.3250	0.2226	1.46	0.158
Equity / Total liabilities (pp)	−0.0006	0.0250	−0.02	0.982
GDP growth (pp)	0.0848	0.0183	4.64	0.000***
Inflation, CPI (pp)	0.1306	0.0314	4.16	0.000***
Unemployment rate (pp)	0.3149	0.1249	2.52	0.019**
Constant	−0.6280	1.3232	−0.47	0.640

Source 10: Author's Own Calculation.

For NPL, the overidentification test following the random-effects regression rejects the orthogonality condition (Sargan–Hansen $\chi^2(4)=12.50$, $p=0.014$), indicating correlation between bank effects and regressors. We therefore estimate NPL models using bank fixed effects with cluster-robust standard errors. NPL (bank fixed effects, clustered SEs; 23 banks, 230 bank-years, 2015–2024). The FE model is jointly significant $F(6,22) = 28.68$, $p<0.001$) with strong within-fit ($R^2=0.52$); about 37% of residual variance is between banks (rho=0.37). CRAR = −0.1212 ($p=0.002$) implying roughly 0.7 percentage-point lower NPL for a + 1 pp increase in CRAR around the sample mean NPL. Macroeconomic headwinds raise NPLs: GDP growth (+0.0848, $p<0.001$), inflation (+0.1306, $p<0.001$), and unemployment (+0.3149, $p=0.019$) are all positively associated with NPL.

Chapter 5: Discussion and Recommendations

This study set out to evaluate whether stronger capital buffers (CRAR) improve profitability and asset quality for Bangladeshi private commercial banks, how bank-specific structures (size and equity/liabilities) condition that link, and how macroeconomic conditions shape bank performance in stressed periods. Using a balanced ten-year panel (2015–2024) of 23 PCBs and panel estimators chosen via formal specification tests (RE for ROA/ROE; FE for NPL), three robust messages emerge.

First, capital matters but differently across outcomes. In the ROA model, CRAR is positive and statistically significant (≈ +0.095 percentage points of ROA per +1 pp CRAR), indicating economically meaningful gains in asset profitability from thicker risk-weighted capital cushions. In contrast, the ROE model shows a positive but imprecisely estimated CRAR coefficient; equity returns are noisier and more sensitive to thin equity bases and episodic losses, so precision is lower. Most notably, in the asset-quality model, CRAR is a strong negative predictor of NPLs (asinh units), consistent with a stability channel whereby better capital attenuates credit risk realization. These patterns align with strands of the literature that find positive capital–profitability links through reduced distress costs (e.g., Akhtar & Lata, 2019; Jalil et al., 2018) while acknowledging potential diminishing returns to "excess capital" for equity returns (Zhao & Tang, 2020; Molyneux & Thornton, 1992). In Bangladesh's current context of elevated NPLs and weakened sectoral CRARs, the finding that CRAR robustly depresses NPLs is policy-salient.

Second, simple leverage (equity/liabilities) supports profitability beyond risk-weighted ratios. The equity-to-liabilities ratio is positively associated with ROA (significant), suggesting that, beyond risk-weights, a thicker un-risk-weighted equity base improves performance—consistent with evidence that higher tangible equity lowers funding costs and tail risk (Köhler, 2012). For ROE, the sign turns negative and is not significant; this is unsurprising because higher equity dilutes the denominator of ROE in the short run. Together, the results imply both risk-weighted (CRAR) and simple capital (equity/liabilities) are informative, but their performance channels differ: CRAR aligns with credit risk containment (lower NPLs) and ROA, while simple leverage tilts the funding/risk mix that supports steadier margins.

Third, macro headwinds weigh on profitability and asset quality. Unemployment is a consistently negative driver of ROA and ROE, while inflation compresses ROA. Interestingly, GDP growth

loads negatively in the RE profitability models. This counter-cyclical sign likely reflects period-specific regimes—e.g., post-COVID rebound, interest-rate policy frictions, pricing caps, or lagged credit losses—that a pure annual growth rate cannot fully disentangle in the presence of limited time variation and no year fixed effects in the preferred RE specification. In the NPL (FE) model, all three macro variables are positively associated with NPLs, consistent with episode-driven stress: higher inflation and unemployment raise default risk, and measured "growth" periods may coincide with portfolio seasoning or reclassification waves. These results echo regional evidence that macro volatility, policy shifts, and credit cycles materially shape banks' loss dynamics (Bikker & Hu, 2002; Hossain & Ahmed, 2016).

5.1 Specific objectives, hypotheses, and findings

5. Objective 1: Quantify CRAR's effect on performance and risk

H1: CRAR → ROA/ROE positive. Finding: Supported for ROA (positive, statistically significant); not supported for ROE (positive but imprecise).

H2: CRAR → NPL negative. Finding: Supported (significantly negative in FE model).

Interpretation: Stronger risk-weighted capital buffers raise asset profitability and curb credit risk; the CRAR–ROE link is directionally positive but not precise, reflecting ROE volatility/denominator effects.

Objective 2: Assess whether bank structure conditions the capital–profitability link

H3: Size (log assets) → ROA/ROE positive (and/or strengthens CRAR's effect). Finding: Not supported size is not statistically significant in preferred models.

H4: Equity-to-liabilities → ROA/ROE positive (and/or strengthens CRAR's effect). Finding: Partially supported positive and significant for ROA; not significant for ROE (and not for NPL).

Interpretation: Simple (un-risk-weighted) capital supports ROA beyond CRAR; scale does not systematically improve profitability in this sample. Evidence of moderation is limited; direct effects dominate.

Objective 3: Identify macroeconomic influences on profitability/NPLs, especially in stress

H5: GDP growth → ROA/ROE positive. Finding: Not supported in preferred RE profitability models (negative sign, likely period/regime effects); GDP is positively associated with NPL in the FE specification.

H6: Inflation → ROA/ROE negative. Finding: Supported for ROA; marginal for ROE (negative, weakly significant).

H7: Unemployment → ROA/ROE negative. Finding: Supported (significantly negative).

Interpretation: Macro headwinds—especially inflation and unemployment—compress profitability; GDP's counter-intuitive sign likely reflects policy/regulatory episodes and limited T.

Objective 4: Profile direct bank-specific determinants (net of CRAR and macros)

Finding: Equity-to-liabilities is positively associated with ROA; size is not significant; neither variable robustly predicts NPL once bank FE absorb time-invariant heterogeneity.

5.2 Policy recommendations

1. Targeted recapitalization and credible buffers. Prioritize restoring CRAR for under-capitalized PCBs, with conditional capital injections and time-bound remedial plans under PCA; calibrate countercyclical buffers as conditions normalize.
2. NPL resolution at scale. Pair higher capital with accelerated NPL cleanup: tighter provisioning, realistic collateral valuation, time-bound write-off policies, and (if necessary) an asset-management vehicle for legacy problem loans.
3. Strengthen simple capital metrics. Because equity/liabilities improve ROA, require transparent disclosure of leverage-based metrics alongside CRAR; discourage dividend payouts until both ratios meet thresholds.
4. Risk-based pricing and margin flexibility. Allow pricing to reflect borrower risk and funding costs to avoid inflation-induced margin compression; ensure pass-through of policy rates and improve interest-rate risk management.
5. Supervisory intensity and governance. Intensify on-site inspections for high-NPL, low-CRAR banks; enforce fit-and-proper criteria, related-party lending limits, and board accountability.
6. Stress testing and early warning. Institutionalize quarterly, bank-level stress tests linked to PCA triggers; publish summary results to reinforce market discipline.
7. Data quality and IFRS 9 discipline. Enhance credit registry coverage, obligor connect-ups, and forward-looking expected-credit-loss (ECL) practices to prevent "evergreening".

Collectively, these measures link the study's micro findings (capital and leverage improve ROA; CRAR reduces NPLs) to macroprudential levers that can stabilize profitability during shocks.

Chapter 6: Conclusion

This study examined whether and how capital strength translates into profitability and better asset quality for Bangladeshi private commercial banks during a decade marked by rising NPLs and capital strain. Using a 2015–2024 panel of 23 PCBs, we estimated random-effects models for profitability (ROA, ROE) and a fixed-effects model for asset quality (NPL), with estimator choice validated by Hausman tests and inference made robust to serial correlation via bank-clustered standard errors. Three main conclusions stand out. First, capital buffers matter. Higher CRAR is associated with higher ROA and lower NPLs, confirming the resilience channel: well-capitalized banks absorb shocks and sustain margins. For ROE, the positive CRAR coefficient lacks statistical precision, reflecting ROE's volatility and denominator sensitivity. Second, simple leverage complements risk-weighted capital. Together, the results indicate that rebuilding capital and accelerating NPL resolution are necessary complements, not substitutes, for restoring sustainable profitability. The study contributes Bangladesh-specific evidence through end-2024, during a period of PCA enforcement, merger discussions, and interest-rate regime changes. Methodologically, it pairs risk-weighted and simple capital measures, tests an asset-quality outcome (NPL) directly, and documents diagnostics (autocorrelation, multicollinearity) to support credible inference. Limitations include reliance on secondary disclosures, potential measurement inconsistencies in NPL classification, and limited variation in macro series (T=10), which constrains identification in models with year effects. Endogeneity concerns (profits ↔ capital) remain plausible; while our specifications and robustness (clustering, estimator tests) mitigate bias, a dynamic GMM design could better address simultaneity. Future research should (i) estimate dynamic panel models (e.g., system GMM) that treat CRAR and profitability as jointly determined; (ii) explore non-linear/threshold effects of CRAR (e.g., benefits above regulatory floors); (iii) dissect portfolio composition (sectoral exposures, collateral, maturity) and governance variables (ownership, board independence) as moderators; (iv) incorporate market-based risk measures (funding spreads, stock-implied volatility for listed banks); and (v) run event-study analyses around key regulatory changes. Such extensions would refine the capital–profitability–risk nexus and sharpen policy design to restore trust and resilience in Bangladesh's banking system. In sum, the evidence supports a clear policy message: rebuild capital, resolve NPLs, and restore risk-sensitive pricing. Doing so simultaneously strengthens profitability (ROA), stabilizes asset quality (NPL), and lays the foundation for durable financial intermediation in Bangladesh.

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Appendix

Appendix A: Panel Data For 23 Private Commercial Banks

banks	year	roa	roe	npl	crar	total_asse	equity_to_	gdp_grow	inflation	banks	year	roa	roe	npl	crar	total_asse	equity_to	gdp_grow	inflation
AB Bank PLC	2015	0.48	6.03	3.16	11.09	5.45486	7.99691	6.6	6.19428	Dhaka Bar	2024	0.31	5.71	5.33	13.41	5.63494	5.38272	4.2	10.4657
AB Bank PLC	2016	0.44	5.68	5.19	10.79	5.49808	7.34792	7.1	5.51353	Dutch-Bar	2015	1.3	19.3	3.7	13.7	5.38749	9.2993	6.6	6.19428
AB Bank PLC	2017	0.01	0.13	7.15	10.8	5.49771	7.23265	6.6	5.70207	Dutch-Bar	2016	0.7	10.3	5.2	13.1	5.42291	8.17133	7.1	5.51353
AB Bank PLC	2018	0.01	0.08	33.07	10.03	5.50856	7.02269	7.3	5.54362	Dutch-Bar	2017	0.9	13.2	4.7	13	5.49402	8.92767	6.6	5.70207
AB Bank PLC	2019	0.05	0.74	18.28	10.12	5.56297	6.24779	7.9	5.592	Dutch-Bar	2018	1.9	19.7	4.1	15.6	5.53966	10.5392	7.3	5.54362
AB Bank PLC	2020	0.1	1.65	16.79	10.84	5.58668	6.34237	3.4	5.69107	Dutch-Bar	2019	1.2	17.2	4.4	15.5	5.59147	10.8386	7.9	5.592
AB Bank PLC	2021	0.16	2.63	14.15	10.74	5.61312	5.92396	6.9	5.54565	Dutch-Bar	2020	1.3	18.4	2.2	17.2	5.67427	10.4713	3.4	5.69107
AB Bank PLC	2022	0.17	2.78	20.23	10.31	5.61181	5.97089	7.1	7.69695	Dutch-Bar	2021	1.1	16.1	3.7	16.41	5.7113	10.5218	6.9	5.54565
AB Bank PLC	2023	0.17	2.95	30	10.26	5.63777	5.63438	5.8	9.8835	Dutch-Bar	2022	1.1	14.4	4.3	15.6	5.74466	9.99352	7.1	7.69695
AB Bank PLC	2024	-4.54	-131.14	67.14	3.47	5.61265	1.1662	4.2	10.4657	Dutch-Bar	2023	1.4	17.9	4.1	16.16	5.7737	10.8294	5.8	9.8835
BRAC Bank	2015	1.13	13.32	5.99	12.23	5.3512	9.14784	6.6	6.19428	Dutch-Bar	2024	0.74	9.5	7.7	13.83	5.83243	8.76011	4.2	10.4657
BRAC Bank	2016	1.89	22.16	3.4	12.26	5.39551	9.43856	7.1	5.51353	Eastern Ba	2015	1.23	10.95	3.27	14.24	5.27775	12.123	6.6	6.19428
BRAC Bank	2017	2.02	22.14	3.56	12.72	5.46165	9.86338	6.6	5.70207	Eastern Ba	2016	1.33	12.94	2.69	15.1	5.32466	10.7923	7.1	5.51353
BRAC Bank	2018	1.87	19.25	3.1	13.67	5.49889	11.1488	7.3	5.54362	Eastern Ba	2017	1.04	11.41	2.5	14.09	5.40324	9.32493	6.6	5.70207
BRAC Bank	2019	1.65	15.6	3.99	15.07	5.56575	11.4025	7.9	5.592	Eastern Ba	2018	1.15	13.83	2.35	12.16	5.45094	8.85072	7.3	5.54362
BRAC Bank	2020	1.19	10.69	2.93	14.55	5.59877	13.26	3.4	5.69107	Eastern Ba	2019	1.3	16.52	3.35	14.74	5.52526	8.25834	7.9	5.592
BRAC Bank	2021	1.31	11	3.9	14.36	5.65233	13.7887	6.9	5.54565	Eastern Ba	2020	1.22	15.04	2.72	15.23	5.52755	9.40888	3.4	5.69107
BRAC Bank	2022	1.14	10.22	3.72	14.04	5.75069	11.5661	7.1	7.69695	Eastern Ba	2021	1.28	15.51	3.7	14.08	5.58974	8.66874	6.9	5.54565
BRAC Bank	2023	1.13	11.95	3.38	14	5.86137	9.62107	5.8	9.8835	Eastern Ba	2022	1.21	15.46	2.78	14.61	5.59047	9.75467	7.1	7.69695
BRAC Bank	2024	1.46	17.43	2.63	14.9	5.96854	8.91666	4.2	10.4657	Eastern Ba	2023	1.27	16.85	3.1	15.07	5.70634	8.48306	5.8	9.8835
Bank Asia	2015	1.26	14.36	4.26	12.46	5.35092	9.24146	6.6	6.19428	Eastern Ba	2024	1.34	18.63	3.34	15.11	5.78962	7.51572	4.2	10.4657
Bank Asia	2016	0.65	8.13	5.41	12.42	5.40346	8.06688	7.1	5.51353	IFIC Bank	2015	0.53	7.96	6.46	10.07	4.06647	6.99184	6.6	6.19428
Bank Asia	2017	0.76	10.21	4.38	14.89	5.46089	7.85767	6.6	5.70207	IFIC Bank	2016	0.65	9.91	5.29	11.25	4.10856	6.97031	7.1	5.51353
Bank Asia	2018	0.76	10.21	4.1	15.05	5.48755	8.21973	7.3	5.54362	IFIC Bank	2017	0.92	12.39	6.4	12.57	5.40355	8.82601	6.6	5.70207
Bank Asia	2019	0.59	8.16	4.61	17.93	5.54876	7.52002	7.9	5.592	IFIC Bank	2018	0.59	7.4	6.16	12.63	5.4519	8.47496	7.3	5.54362
Bank Asia	2020	0.53	7.81	3.24	17.16	5.61168	7.17571	3.4	5.69107	IFIC Bank	2019	0.81	10.46	5.37	12.8	5.50099	8.41346	7.9	5.592
Bank Asia	2021	0.64	9.89	5.14	15.72	5.64176	6.65724	6.9	5.54565	IFIC Bank	2020	0.17	2.26	3.97	12.24	5.55242	7.59901	3.4	5.69107
Bank Asia	2022	0.67	10.88	4.87	17.7	5.67062	6.51264	7.1	7.69695	IFIC Bank	2021	0.52	7.55	6.09	13.1	5.6084	7.16053	6.9	5.54565
Bank Asia	2023	0.51	8.24	6.7	16.45	5.67599	6.63805	5.8	9.8835	IFIC Bank	2022	0.68	10.38	5.61	12.78	5.66771	6.91458	7.1	7.69695
Bank Asia	2024	0.78	9.23	11.4	15.61	5.71291	6.37866	4.2	10.4657	IFIC Bank	2023	0.51	14.7	7.99	11.82	5.71945	6.52225	5.8	9.8835
City Bank	2015	1.9	18.3	7.6	13.2	5.30583	11.15	6.6	6.19428	IFIC Bank	2024	-0.19	-0.02	54.97	7.27	5.75258	5.80662	4.2	10.4657
City Bank	2016	1.7	18.9	6	13.2	5.40616	8.82	7.1	5.51353	Jamuna Ba	2015	1.16	12.39	6.92	12.83	5.15491	12.3541	6.6	6.19428
City Bank	2017	1.4	18	5.3	14.7	5.44017	9.92	6.6	5.70207	Jamuna Ba	2016	1.21	11.89	4.05	10.93	5.22639	10.3878	7.1	5.51353
City Bank	2018	0.7	13.5	3.5	13.4	5.51159	8.13	7.3	5.54362	Jamuna Ba	2017	1.11	12.92	4.02	11.86	5.2946	8.49737	6.6	5.70207
City Bank	2019	0.7	9.9	5.8	15.2	5.54985	7.72	7.9	5.592	Jamuna Ba	2018	1.1	13.83	3.77	13.58	5.35222	8.73349	7.3	5.54362
City Bank	2020	1.1	14.8	4	15.5	5.58311	8.14	3.4	5.69107	Jamuna Ba	2019	1.11	14.8	3.7	14.25	5.38548	7.60099	7.9	5.592
City Bank	2021	1.2	15.8	4.9	14.2	5.62003	8.1	6.9	5.54565	Jamuna Ba	2020	1.1	12.93	2.95	15.5	5.38375	11.1067	3.4	5.69107
City Bank	2022	1	14.1	3.9	14.46	5.70488	6.93	7.1	7.69695	Jamuna Ba	2021	0.98	10.88	2.97	16.42	5.42295	8.79569	6.9	5.54565
City Bank	2023	1.2	17.6	3.6	15.8	5.74487	7.18	5.8	9.8835	Jamuna Ba	2022	0.58	7.69	5.32	16.75	5.45194	7.51223	7.1	7.69695
City Bank	2024	1.7	26.1	3.7	16	5.84345	7.03	4.2	10.4657	Jamuna Ba	2023	0.8	11.85	4.96	16.64	5.48765	7.018	5.8	9.8835
Dhaka Bank	2015	0.86	10.74	4.66	10.46	5.24641	8.21547	6.6	6.19428	Jamuna Ba	2024	0.83	13.33	6.92	16.37	5.56406	6.32735	4.2	10.4657
Dhaka Bank	2016	0.77	10.15	4.01	13.67	5.30576	7.69444	7.1	5.51353	Meghna B	2015	2.2	23.18	0	27.23	4.35774	28.5139	6.6	6.19428
Dhaka Bank	2017	0.69	9.21	5.98	11.96	5.36069	7.11691	6.6	5.70207	Meghna B	2016	2.1	17.6	0.99	23.55	4.50299	20.4418	7.1	5.51353
Dhaka Bank	2018	0.54	8.18	4.99	13.84	5.43771	6.45633	7.3	5.54362	Meghna B	2017	1.18	14.52	3	19.52	4.59962	15.718	6.6	5.70207
Dhaka Bank	2019	0.56	9.29	4.74	16.12	5.45486	6.42686	7.9	5.592	Meghna B	2018	0.22	11.87	6	11.875	4.64196	14.3915	7.3	5.54362
Dhaka Bank	2020	0.7	11.28	3.13	14.52	5.47032	6.79528	3.4	5.69107	Meghna B	2019	0.29	8.31	7.41	17.89	4.69325	13.0159	7.9	5.592
Dhaka Bank	2021	0.65	10.53	3.32	14.65	5.5255	6.42381	6.9	5.54565	Meghna B	2020	0.98	14.29	6.8	17.42	4.72322	13.5984	3.4	5.69107
Dhaka Bank	2022	0.49	8.09	5.08	14.12	5.53977	6.37633	7.1	7.69695	Meghna B	2021	0.52	11.26	5.6	17.12	4.81481	10.4804	6.9	5.54565
Dhaka Bank	2023	0.48	8.14	4.88	15.04	5.57823	6.10503	5.8	9.8835	Meghna B	2022	0.29	7.66	6.73	21.3	4.8098	11.2709	7.1	7.69695

banks	year	roa	roe	npl	crar	total_asse	equity_to_	gdp_grow	inflation	banks	year	roa	roe	npl	crar	total_asse	equity_to	gdp_grow	inflation
Meghna Bank	2023	1.05	11.24	4.53	18.41	4.92143	9.59498	5.8	9.8835	NRB Bank	2021	0.73	6.68	3.39	18.23	4.78633	12.9336	6.9	5.54565
Meghna Bank	2024	0.79	6.4	4.78	15.06	5.03995	7.05602	4.2	10.4657	NRB Bank	2022	0.82	7.28	3.22	16.41	4.86482	11.859	7.1	7.69695
Mercantile Bank	2015	0.79	9.6	4.95	11.87	5.26198	9.27378	6.6	6.19428	NRB Bank	2023	1	9.98	4.98	15.44	4.93288	10.5021	5.8	9.8835
Mercantile Bank	2016	1.15	13.78	5.13	13.03	5.3099	8.97388	7.1	5.51353	NRB Bank	2024	0.08	8.5	15.61	14.54	4.99182	9.78	4.2	10.4657
Mercantile Bank	2017	1.3	17.55	3.79	11.93	5.41526	7.2486	6.6	5.70207	NRBC Ban	2015	1.83	11.55	0.27	18.93	4.55942	17.44	6.6	6.19428
Mercantile Bank	2018	1.09	16.55	4.82	13.28	5.46447	6.85127	7.3	5.54362	NRBC Ban	2016	1.93	15.97	0.52	13.5	4.72932	12.9357	7.1	5.51353
Mercantile Bank	2019	0.78	10.99	4.86	13.92	5.50019	7.07664	7.9	5.592	NRBC Ban	2017	1.67	15.3	2.46	13.58	4.75989	13.2705	6.6	5.70207
Mercantile Bank	2020	0.73	10.05	4.72	13.61	5.51955	7.16003	3.4	5.69107	NRBC Ban	2018	1.5	13.89	2.94	14.03	4.827	12.2832	7.3	5.54362
Mercantile Bank	2021	0.99	14.7	4.54	14.09	5.55559	7.28459	6.9	5.54565	NRBC Ban	2019	1.47	15.04	3.2	13.4	4.98838	9.99651	7.9	5.592
Mercantile Bank	2022	0.59	8.87	7.09	14.35	5.58244	7.07882	7.1	7.69695	NRBC Ban	2020	1.34	15.28	2.93	12.52	5.06686	9.17678	3.4	5.69107
Mercantile Bank	2023	0.52	7.92	6.08	14.79	5.60441	6.90731	5.8	9.8835	NRBC Ban	2021	1.54	19.29	4.56	13.45	5.18875	8.29333	6.9	5.54565
Mercantile Bank	2024	0.15	2.44	17.25	13.07	5.64806	6.16168	4.2	10.4657	NRBC Ban	2022	0.98	14.06	4.69	12.44	5.3018	6.90178	7.1	7.69695
Midland Bank	2015	1.84	8.63	1.01	26.29	4.37827	26.92	6.6	6.19428	NRBC Ban	2023	0.88	13.62	5.35	13.7	5.33844	6.95761	5.8	9.8835
Midland Bank	2016	2.26	12.77	0.68	23.3	4.51959	18.02	7.1	5.51353	NRBC Ban	2024	0.47	7.93	15.79	12	5.37447	6.70842	4.2	10.4657
Midland Bank	2017	1.65	13.02	1.68	21.03	4.62272	16.21	6.6	5.70207	One Bank	2015	1.39	7.86	3.58	10.93	5.18806	8.11918	6.6	6.19428
Midland Bank	2018	1.51	11.69	1.2	19.33	4.6604	16.55	7.3	5.54362	One Bank	2016	1.2	7.58	4.93	13.89	5.27471	7.36689	7.1	5.51353
Midland Bank	2019	1.05	8.56	2.69	16.06	4.76713	12.58	7.9	5.592	One Bank	2017	1.05	8.26	5.31	11.56	5.35641	6.62108	6.6	5.70207
Midland Bank	2020	1.13	8.96	1.16	17.44	4.80035	13.71	3.4	5.69107	One Bank	2018	0.56	6.88	7.05	11.93	5.42496	5.71889	7.3	5.54362
Midland Bank	2021	0.78	6.85	3.17	15.87	4.83897	12.02	6.9	5.54565	One Bank	2019	0.59	6	9.24	12.8	5.47403	5.69581	7.9	5.592
Midland Bank	2022	0.75	7.47	2.78	14.83	4.90237	10.59	7.1	7.69695	One Bank	2020	0.48	7.62	8.6	13.02	5.4864	6.88139	3.4	5.69107
Midland Bank	2023	1.39	13.49	3.81	17.28	4.9213	12.32	5.8	9.8835	One Bank	2021	0.34	5.2	11.55	12.03	5.49614	7.01146	6.9	5.54565
Midland Bank	2024	0.88	8.69	3.78	16.78	5.01188	10.39	4.2	10.4657	One Bank	2022	0.65	9.62	13.97	11.77	5.5035	7.4401	7.1	7.69695
Modhumoti Bank	2015	3.81	14.67	0	35.26	4.37596	30.2906	6.6	6.19428	One Bank	2023	0.36	5.6	11.95	13.09	5.50729	7.60114	5.8	9.8835
Modhumoti Bank	2016	1.65	8.43	0.26	19.42	4.51088	20.1601	7.1	5.51353	One Bank	2024	0.39	5.54	17.47	13.6	5.52907	7.45364	4.2	10.4657
Modhumoti Bank	2017	1.93	13.36	0.29	16.76	4.66063	14.3118	6.6	5.70207	Prime Bar	2015	0.8	8.41	7.82	12.74	5.42703	10.96	6.6	6.19428
Modhumoti Bank	2018	2.02	16.38	1.83	15.04	4.71343	13.771	7.3	5.54362	Prime Bar	2016	0.81	8.49	5.96	12.45	5.43493	10.24	7.1	5.51353
Modhumoti Bank	2019	1.29	12.31	2.53	13.92	4.85203	10.2283	7.9	5.592	Prime Bar	2017	0.38	4.24	5.45	14.01	5.44913	9.63	6.6	5.70207
Modhumoti Bank	2020	2.28	21.7	1.22	17.84	4.86282	13.3012	3.4	5.69107	Prime Bar	2018	0.76	8.6	6.16	17.04	5.4682	9.78	7.3	5.54362
Modhumoti Bank	2021	1.48	13.38	2.84	18.05	4.90068	11.6229	6.9	5.54565	Prime Bar	2019	0.65	7.57	4.8	17.42	5.50819	8.84	7.9	5.592
Modhumoti Bank	2022	1.18	11.94	1.7	16.07	4.95725	10.4213	7.1	7.69695	Prime Bar	2020	0.64	7.7	4.84	17.28	5.59223	7.35	3.4	5.69107
Modhumoti Bank	2023	1.18	12.56	1.67	16.08	4.98487	10.293	5.8	9.8835	Prime Bar	2021	0.97	11.57	5.78	17.17	5.65198	6.66	6.9	5.54565
Modhumoti Bank	2024	1.28	13.63	2.3	15.8	4.99635	10.5225	4.2	10.4657	Prime Bar	2022	0.86	12.07	5.06	17.28	5.67872	7.18	7.1	7.69695
Mutual Trust Ban	2015	1.04	17.4	2.08	12.02	5.16457	6.51149	6.6	6.19428	Prime Bar	2023	1.08	15.44	4.65	17.17	5.67872	7.69	5.8	9.8835
Mutual Trust Ban	2016	0.94	15.61	4.36	11.29	5.21846	6.31187	7.1	5.51353	Prime Bar	2024	1.3	17.36	4.27	17.19	5.7096	8.3	4.2	10.4657
Mutual Trust Ban	2017	1.08	18.35	4.3	13.76	5.30482	6.19077	6.6	5.70207	SBAC Banl	2015	1.18	7.85	0.1	21.19	4.56452	13.3019	6.6	6.19428
Mutual Trust Ban	2018	0.82	13.83	5.39	12.86	5.34726	6.37238	7.3	5.54362	SBAC Banl	2016	2.31	18.09	0	18.59	4.65896	12.3165	7.1	5.51353
Mutual Trust Ban	2019	0.56	9.03	5.39	12.91	5.40979	6.76992	7.9	5.592	SBAC Banl	2017	1.8	16.16	0.87	16.54	4.77042	11.0942	6.6	5.70207
Mutual Trust Ban	2020	0.37	5.83	4.61	12.92	5.43019	6.74349	3.4	5.69107	SBAC Banl	2018	1.36	12.4	1.97	15.88	4.84918	10.5145	7.3	5.54362
Mutual Trust Ban	2021	1.03	16.22	5.8	14.16	5.48643	6.67954	6.9	5.54565	SBAC Banl	2019	1.25	12.55	6.09	14.73	4.92719	9.8895	7.9	5.592
Mutual Trust Ban	2022	0.72	11.56	5.77	14.48	5.54856	6.29198	7.1	7.69695	SBAC Banl	2020	1.1	10.78	6.22	13.45	4.94647	10.5275	3.4	5.69107
Mutual Trust Ban	2023	0.75	8.81	6.6	14.98	5.5612	6.5752	5.8	9.8835	SBAC Banl	2021	0.57	5.36	5.85	13.91	4.9904	10.866	6.9	5.54565
Mutual Trust Ban	2024	0.76	11.78	6.95	13.62	5.65735	5.57849	4.2	10.4657	SBAC Banl	2022	0.58	5.5	5.18	13.78	5.02656	10.1871	7.1	7.69695
NRB Bank	2015	2.02	5.27	0.2	26.11	4.31249	27.893	6.6	6.19428	SBAC Banl	2023	0.49	4.98	5.94	13.82	5.06917	9.48743	5.8	9.8835
NRB Bank	2016	3.18	11.02	1.9	27.44	4.39322	25.7188	7.1	5.51353	SBAC Banl	2024	0.09	0.99	16.13	13.75	5.09552	8.89972	4.2	10.4657
NRB Bank	2017	1.48	9.11	2.46	18.4	4.54332	15.4254	6.6	5.70207	Southeast	2015	1.23	11.86	4.25	11.52	5.41617	6.93078	6.6	6.19428
NRB Bank	2018	1.47	11.74	3.71	17.22	4.64491	13.501	7.3	5.54362	Southeast	2016	0.88	9.06	4.89	12.15	5.46508	6.48114	7.1	5.51353
NRB Bank	2019	-0.25	-2.42	12	13.32	4.73817	10.3398	7.9	5.592	Southeast	2017	0.37	4.46	5.99	10.84	5.53057	8.72796	6.6	5.70207
NRB Bank	2020	1.22	12.28	3.71	16.07	4.74322	11.7113	3.4	5.69107	Southeast	2018	0.69	9.17	5.87	12.47	5.58158	8.50893	7.3	5.54362

banks	year	roa	roe	npl	crar	total_asse	equity_to_	gdp_grow	inflation
Southeast Bank	2019	0.62	8.56	4.87	11.61	5.62563	8.3242	7.9	5.592
Southeast Bank	2020	0.47	6.91	3.1	14.52	5.66773	6.82367	3.4	5.69107
Southeast Bank	2021	0.37	5.98	4.81	14.02	5.69416	6.44199	6.9	5.54565
Southeast Bank	2022	0.38	5.77	6.54	12.88	5.69197	6.65234	7.1	7.69695
Southeast Bank	2023	0.42	6.8	9.65	13.52	5.7074	9.56519	5.8	9.8835
Southeast Bank	2024	0.09	1.19	14.82	11.71	5.73572	8.07615	4.2	10.4657
Trust Bank PLC	2015	0.95	17.45	2.74	10.84	5.25583	5.57181	6.6	6.19428
Trust Bank PLC	2016	1.03	19.42	3.21	14.7	5.32272	5.61379	7.1	5.51353
Trust Bank PLC	2017	0.77	14.86	3.35	12.95	5.3798	5.32033	6.6	5.70207
Trust Bank PLC	2018	0.76	15.14	7.9	13.87	5.41437	5.22292	7.3	5.54362
Trust Bank PLC	2019	0.73	14.6	5.49	14.35	5.47196	5.45771	7.9	5.592
Trust Bank PLC	2020	0.55	11.18	4.51	13.73	5.55201	5.04502	3.4	5.69107
Trust Bank PLC	2021	0.76	15.55	3.65	14.14	5.5629	5.25814	6.9	5.54565
Trust Bank PLC	2022	0.76	15.7	4.76	13.73	5.62108	4.98563	7.1	7.69695
Trust Bank PLC	2023	0.99	20.2	6.08	13.75	5.65689	5.3132	5.8	9.8835
Trust Bank PLC	2024	0.75	15.82	6.36	13.21	5.74615	4.73188	4.2	10.4657
United Commerc	2015	0.57	9.9	5.23	12.16	5.46812	9.53895	6.6	6.19428
United Commerc	2016	0.67	11.28	8.01	11.53	5.51815	8.47334	7.1	5.51353
United Commerc	2017	0.61	9.05	7.38	12.07	5.56028	7.87102	6.6	5.70207
United Commerc	2018	0.61	9.69	6.79	12.9	5.60503	8.01933	7.3	5.54362
United Commerc	2019	0.49	7.56	3.63	14.68	5.65007	7.91904	7.9	5.592
United Commerc	2020	0.61	8.42	2.55	14.92	5.70994	7.49426	3.4	5.69107
United Commerc	2021	0.6	8.76	4.41	13.64	5.71524	6.6325	6.9	5.54565
United Commerc	2022	0.67	10.27	5.99	13.06	5.79801	6.976	7.1	7.69695
United Commerc	2023	0.41	6.41	5.5	12.58	5.8357	6.67194	5.8	9.8835
United Commerc	2024	0.08	1.44	14.9	10.59	5.8858	5.74851	4.2	10.4657
Uttara Bank	2015	0.99	11.32	8.27	12.61	5.18034	8.71819	6.6	6.19428
Uttara Bank	2016	0.94	12.9	7.75	13.38	5.21063	8.3728	7.1	5.51353
Uttara Bank	2017	0.87	8.67	6.75	12.92	5.25256	7.92239	6.6	5.70207
Uttara Bank	2018	0.9	9.84	6.33	12.49	5.27882	7.84542	7.3	5.54362
Uttara Bank	2019	0.97	9.44	7.81	13	5.28592	8.19948	7.9	5.592
Uttara Bank	2020	0.96	9.6	6.36	14.01	5.34753	8.09434	3.4	5.69107
Uttara Bank	2021	0.85	7.08	7.53	15.11	5.37954	8.22028	6.9	5.54565
Uttara Bank	2022	1.1	8.27	6.83	15.36	5.39033	8.91536	7.1	7.69695
Uttara Bank	2023	1.22	7.97	5.46	15.5	5.42031	9.32811	5.8	9.8835
Uttara Bank	2024	1.74	7.77	5.62	16.31	5.44878	10.1558	4.2	10.4657

Appendix B: Hausman Test for Dependent Variables

A.B.1: Hausman Test for ROA

```
                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |       fe           re         Difference          S.E.
-------------+----------------------------------------------------------------
        crar |    .0975848     .0947963        .0027885        .0050146
 total_asset |     .047565      .110583       -.0630179        .1259574
equity_to_~s |    .0316245     .0364227       -.0047981        .0051427
  gdp_growth |   -.0585513    -.0592944        .0007431        .0073487
   inflation |    -.094147    -.0963712        .0022242        .0110334
unemployme~e |     -.38534    -.3920779        .0067378        .0422664
------------------------------------------------------------------------------
                           b = consistent under Ho and Ha; obtained from xtreg
            B = inconsistent under Ha, efficient under Ho; obtained from xtreg

    Test:  Ho:  difference in coefficients not systematic

                  chi2(6) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =        1.13
                Prob>chi2 =      0.9803
```

A.B.2: Hausman Test for ROE

```
                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |       fe           re         Difference          S.E.
-------------+----------------------------------------------------------------
        crar |     1.24518     1.384775       -.1395951        .1595296
 total_asset |    6.772771     4.651117        2.121654        3.498238
equity_to_~s |   -.3524843    -.3625146        .0100303        .1638532
  gdp_growth |   -.4648016    -.3654919       -.0993097        .1975825
   inflation |   -1.723514    -1.541465       -.1820493        .3045754
unemployme~e |    -4.94457    -4.324824        -.619745        1.165563
------------------------------------------------------------------------------
                           b = consistent under Ho and Ha; obtained from xtreg
            B = inconsistent under Ha, efficient under Ho; obtained from xtreg

    Test:  Ho:  difference in coefficients not systematic

                  chi2(6) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =        2.11
                Prob>chi2 =      0.9093
```

A.B.3: Hausman Test for NPL

	Coefficients (b) fe	(B) re	(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
crar	-1.154159	-1.226675	.0725163	.0809912
total_asset	.9204501	.6273469	.2931033	1.850622
equity_to_~s	.5347305	.4892442	.0454863	.0832461
gdp_growth	.2597561	.2561321	.003624	.1018847
inflation	1.401709	1.391589	.0101204	.1604924
unemployme~e	2.244195	2.236753	.0074423	.6114712

b = consistent under Ho and Ha; obtained from xtreg
B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

chi2(3) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 5.02
Prob>chi2 = 0.1703
(V_b-V_B is not positive definite)

Fixed-effects (within) regression — Number of obs = 230
Group variable: bank_id — Number of groups = 23

R-sq: within = 0.5177 — Obs per group: min = 10
between = 0.5912 — avg = 10.0
overall = 0.5494 — max = 10

F(6,22) = 28.68
corr(u_i, Xb) = -0.0009 — Prob > F = 0.0000

(Std. Err. adjusted for 23 clusters in bank_id)

npl_asinh	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
crar	-.1212329	.0334571	-3.62	0.002	-.1906186	-.0518471
total_asset	.3249889	.2226024	1.46	0.158	-.1366603	.7866381
equity_to_liabilities	-.0005856	.0250087	-0.02	0.982	-.0524505	.0512793
gdp_growth	.0848035	.0182751	4.64	0.000	.0469034	.1227037
inflation	.1305959	.031384	4.16	0.000	.0655096	.1956823
unemployment_rate	.3148647	.1248781	2.52	0.019	.0558833	.5738461
_cons	-.628023	1.323238	-0.47	0.640	-3.372251	2.116205
sigma_u	.30680879					
sigma_e	.4002362					
rho	.37012973	(fraction of variance due to u_i)				

Appendix C: Stata Regression Output For ROA, ROE and NPL

A.C.1: ROA (RE-GLS, clustered)

```
Random-effects GLS regression                   Number of obs      =       230
Group variable: bank_id                         Number of groups   =        23

R-sq:  within  = 0.9085                         Obs per group: min =        10
       between = 0.9281                                        avg =      10.0
       overall = 0.9162                                        max =        10

                                                Wald chi2(6)       =    218.35
corr(u_i, X)   = 0 (assumed)                    Prob > chi2        =    0.0000

                                   (Std. Err. adjusted for 23 clusters in bank_id)
------------------------------------------------------------------------------
                     |               Robust
                 roa |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
---------------------+--------------------------------------------------------
                crar |   .0947963   .0306965     3.09   0.002     .0346324    .1549603
           total_asset |    .110583   .1660591     0.67   0.505    -.2148868    .4360528
equity_to_liabilities |   .0364227   .0161927     2.25   0.024     .0046855    .0681598
          gdp growth |  -.0592944   .0248722    -2.38   0.017     -.108043   -.0105457
           inflation |  -.0963712   .0329828    -2.92   0.003    -.1610163    -.031726
    unemployment_rate |  -.3920779   .0994821    -3.94   0.000    -.5870591   -.1970966
               _cons |   1.475276   .9284762     1.59   0.112    -.3445044    3.295056
---------------------+--------------------------------------------------------
             sigma_u |  .29189439
             sigma_e |  .38883884
                 rho |  .36041905   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

A.C.2: ROE (RE-GLS, clustered)

```
Random-effects GLS regression                   Number of obs      =       230
Group variable: bank_id                         Number of groups   =        23

R-sq:  within  = 0.8122                         Obs per group: min =        10
       between = 0.8989                                        avg =      10.0
       overall = 0.8510                                        max =        10

                                                Wald chi2(6)       =     18.69
corr(u_i, X)   = 0 (assumed)                    Prob > chi2        =    0.0983

                                   (Std. Err. adjusted for 23 clusters in bank_id)
------------------------------------------------------------------------------
                     |               Robust
                 roe |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
---------------------+--------------------------------------------------------
                crar |   1.384775   .9666663     1.43   0.152    -.5098559    3.279406
           total_asset |   4.651117   4.008682     1.16   0.246    -3.205755    12.50799
equity_to_liabilities |  -.3625146     .36118    -1.00   0.316    -1.070414    .3453851
          gdp_growth |  -.3654919   .2700069    -1.35   0.176    -.8946957    .1637118
           inflation |  -1.541465   .8548666    -1.80   0.071    -3.216973    .1340427
    unemployment_rate |  -4.324824   1.772263    -2.44   0.015    -7.798397   -.8512522
               _cons |   1.760851   22.27632     0.08   0.937    -41.89993    45.42163
---------------------+--------------------------------------------------------
             sigma_u |  4.0484879
             sigma_e |   9.041317
                 rho |  .16701628   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

A.C.3: NPL (FE with asinh transform, clustered)

```
Fixed-effects (within) regression               Number of obs      =       230
Group variable: bank_id                         Number of groups   =        23

R-sq:  within  = 0.5177                         Obs per group: min =        10
       between = 0.5912                                        avg =      10.0
       overall = 0.5494                                        max =        10

                                                F(6,22)            =     28.68
corr(u_i, Xb)  = -0.0009                        Prob > F           =    0.0000

                                     (Std. Err. adjusted for 23 clusters in bank_id)
------------------------------------------------------------------------------------
                |               Robust
           npl_asinh |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
------------------------------------------------------------------------------------
                crar |  -.1212329   .0334571    -3.62   0.002    -.1906186   -.0518471
           total_asset |   .3249889   .2226024     1.46   0.158    -.1366603    .7866381
equity_to_liabilities |  -.0005856   .0250087    -0.02   0.982    -.0524505    .0512793
           gdp_growth |   .0848035   .0182751     4.64   0.000     .0469034    .1227037
            inflation |   .1305959    .031384     4.16   0.000     .0655096    .1956823
    unemployment_rate |   .3148647   .1248781     2.52   0.019     .0558833    .5738461
                _cons |   -.628023   1.323238    -0.47   0.640    -3.372251    2.116205
------------------------------------------------------------------------------------
              sigma_u |  .30680879
              sigma_e |   .4002362
                  rho |  .37012973   (fraction of variance due to u_i)
```

Appendix A.D1. Wooldridge test for autocorrelation AR (1) in panel residuals

DV	F (1,22)	p-value	Decision
ROA	7.664	0.0112	Reject H_0 → AR (1) present
ROE	12.461	0.0019	Reject H_0 → AR (1) present
NPL	20.939	0.0001	Reject H_0 → AR (1) present

Appendix A.D2. Robustness for error structure

To address detected AR (1), we (i) cluster SEs by bank in all FE/RE models, (ii) estimate FE–AR (1) via xtregar and panel-specific AR (1) GLS model.

Appendix E: Variance inflation factors (VIF) for regressors

Variable	VIF	1/VIF
Equity to Total Liabilities	4.09	0.244
CRAR	2.91	0.344
GDP Growth	2.45	0.408
Unemployment Rate	2.21	0.453
Total Asset (log)	2.09	0.478
Inflation (CPI)	2.00	0.500
Mean VIF	2.62	

Notes. VIFs computed from pooled OLS using the core RHS: crar, total_asset, equity_to_liabilities, gdp_growth, inflation, unemployment rate. Fixed effects / year dummies were not included when calculating VIF (to avoid mechanical collinearity). Rule-of-thumb thresholds: VIF > 10 (severe), > 5 (worth scrutiny). All here are below 5.